

CSE 5830 — Homework 3 Hints

These are hints, not answers. Each worked example uses *different* numbers or a *different* graph from the homework. Read the example, then apply the same steps to your problem.

Problem 1: Noisy-AND

Start from noisy-OR (Lec 4, slides 28–31). Each parent X_i has a “mechanism” variable Z_i , and $\lambda_i = P(Z_i = 1 \mid X_i = 1)$, with $Z_i = 0$ whenever $X_i = 0$. The leak Z_0 is always on, with $P(Z_0 = 1) = \lambda_0$. Then $Y = Z_0 \vee Z_1 \vee \cdots \vee Z_K$ (deterministic OR). So $Y = 0$ only when *every* mechanism fails, which gives a product:

$$P(Y = 0 \mid X_1, \dots, X_K) = (1 - \lambda_0) \prod_{i: X_i=1} (1 - \lambda_i).$$

Worked example (noisy-OR). $K = 2$, $\lambda_0 = 0.1$, $\lambda_1 = 0.7$, $\lambda_2 = 0.4$:

$$P(Y = 0 \mid X_1 = 1, X_2 = 1) = (0.9)(0.3)(0.6) = 0.162,$$

$$P(Y = 0 \mid X_1 = 1, X_2 = 0) = (0.9)(0.3) = 0.27 \quad (X_2 \text{ is off, so it contributes no factor}).$$

Notice the product runs only over the parents that are *on*. That is the “independence of causal mechanisms” assumption.

Going from OR to AND (De Morgan). Your Week 3 notes say noisy-AND is the dual of noisy-OR via De Morgan:

$$\neg(X_1 \wedge \cdots \wedge X_K) = \neg X_1 \vee \cdots \vee \neg X_K.$$

Work through these questions:

1. In a deterministic AND, which single condition on one parent is enough to force $Y = 0$? Compare this with the OR, where one parent being on is enough to force $Y = 1$.
2. So in the noisy-AND, which value of Y plays the role that $Y = 1$ played in noisy-OR? And which value of X_i “activates” the mechanism Z_i ?
3. This means the “all mechanisms fail” product is now a product for $P(Y = _ \mid \cdot)$, and it runs over the parents with $X_i = _$. Write that down, then take the complement to get $P(Y = 0 \mid \cdot)$.
4. What does a leak mean for AND? (Hint: Y can come out *wrong* even when every parent looks favorable.)

Convention warning. Different sources define λ_i for noisy-AND differently. Is it the probability that X_i *succeeds* in contributing to $Y = 1$, or the probability that $X_i = 0$ *succeeds* in forcing $Y = 0$? Write one sentence saying which convention you use. Then check your formula against these limiting cases:

- With no leak and all $\lambda_i = 1$, does it reduce to a deterministic AND?
- When every parent is in its “favorable” state, what remains? (It should involve only the leak.)

Part (b). Before you plug in numbers, ask which parents are on and which are off, and which of them appear in the product under your convention. $\lambda_0 = 0$ tells you something about the leak term. What does that factor become?

Problem 2: Factor product and reduction

Definition. $\psi(X, Y, Z) = \phi_1(X, Y) \phi_2(Y, Z)$: for each full assignment (x, y, z) , take the ϕ_1 row that matches (x, y) and the ϕ_2 row that matches (y, z) , and multiply them. The shared variable Y must have the *same* value in both lookups.

Worked example (Koller & Friedman, Fig. 4.3, trimmed to binary).

$A \quad B \mid \phi_1$			$B \quad C \mid \phi_2$			$A \quad B \quad C \mid \psi$			
1	1	0.5	1	1	0.5	1	1	1	$0.5 \cdot 0.5 = 0.25$
1	2	0.8	1	2	0.7	1	1	2	$0.5 \cdot 0.7 = 0.35$
2	1	0.1	2	1	0.1	1	2	1	$0.8 \cdot 0.1 = 0.08$
2	2	0	2	1	0.1	1	2	2	$0.8 \cdot 0.2 = 0.16$
			2	2	0.2	2	1	1	$0.1 \cdot 0.5 = 0.05$
						2	1	2	$0.1 \cdot 0.7 = 0.07$
						2	2	1	$0 \cdot 0.1 = 0$
						2	2	2	$0 \cdot 0.2 = 0$

Sanity check: $2^3 = 8$ rows, and a factor product does *not* need to sum to 1.

Reduction (Lec 5, “Factor reduction”). To reduce ψ to the context $B = 1$, keep only the rows with $B = 1$ and drop the B column:

$A \quad C \mid \psi[B = 1]$		
1	1	0.25
1	2	0.35
2	1	0.05
2	2	0.07

Normalize or not? Factor reduction by itself does not normalize. If you read $\psi(X, Z \mid Y = 1)$ as a conditional *distribution*, then divide by the sum of the kept rows. In this example the sum is 0.72, so the rows become 0.347, 0.486, 0.069, 0.097. You can show both versions and say which one you mean.

Problem 3: Conditionals in a pairwise MN

Recipe.

1. Write the conditional as a ratio: $p(x_i | \text{rest}) = \frac{p(x_i, \text{rest})}{\sum_{x'_i} p(x'_i, \text{rest})}$.
2. Substitute the factored form. The partition function Z cancels.
3. Any factor that does *not* contain x_i is the same in the numerator and in every term of the denominator, so it cancels.
4. What is left is the factors that touch x_i , normalized over x_i .

Worked example (3-node chain $a - b - c$). $p(a, b, c) = \frac{1}{Z} \phi(a, b) \phi(b, c)$.

$$p(b | a, c) = \frac{\frac{1}{Z} \phi(a, b) \phi(b, c)}{\sum_{b'} \frac{1}{Z} \phi(a, b') \phi(b', c)} = \frac{\phi(a, b) \phi(b, c)}{\sum_{b'} \phi(a, b') \phi(b', c)}.$$

$$p(a | b, c) = \frac{\phi(a, b) \cancel{\phi(b, c)}}{\sum_{a'} \phi(a', b) \cancel{\phi(b, c)}} = \frac{\phi(a, b)}{\sum_{a'} \phi(a', b)}.$$

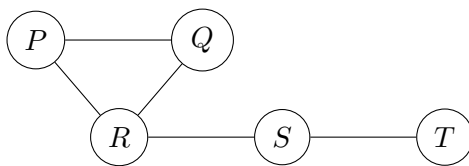
Notice that $p(a | b, c)$ does not depend on c , because c is not a neighbor of a .

For your 4-cycle. For each x_i , ask: which two factors contain x_i ? Which two variables are its neighbors? Are you conditioning on exactly those neighbors? (Compare your answer with the local Markov property in Problem 4.)

Problem 4: Cliques, Gibbs, local Markov, separation

Definitions (Lec 5). A *clique* is a fully connected subset of nodes. A *maximal clique* is a clique that is not contained in a larger clique. A distribution *factorizes over* H if it is a product of factors over cliques of H . It is enough to use one factor per maximal clique, since smaller factors can be multiplied into it.

Worked example graph.



- **Cliques:** singletons; the edges PQ, QR, PR, RS, ST ; the triangle PQR . (Check: is QRS a clique? No, because there is no $Q-S$ edge.)
- **Maximal cliques:** $\{P, Q, R\}$, $\{R, S\}$, $\{S, T\}$, so $P = \frac{1}{Z} \phi_1(P, Q, R) \phi_2(R, S) \phi_3(S, T)$.
- **Pairwise MN:** one factor per *edge*, even inside the triangle: $P = \frac{1}{Z} \phi(P, Q) \phi(Q, R) \phi(P, R) \phi(R, S) \phi(S, T)$, with $Z = \sum_{p,q,r,s,t} (\text{same product})$.

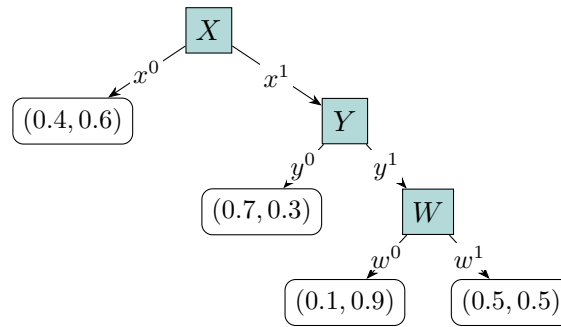
- **Local Markov:** each node is independent of everything else given its neighbors. For example, $P \perp \{S, T\} \mid \{Q, R\}$ and $R \perp T \mid \{P, Q, S\}$. Do this for every node.
- **Separation:** Is $P \perp T \mid S$? Every path from P to T goes through S , so yes. Is $P \perp T \mid Q$? No, because the path $P - R - S - T$ avoids Q .

For your graph. First list every edge carefully. The diagonal $B-F$ edge matters: which triangles does it create? Check each triple $\{B, ?, F\}$. For (d), look for *any* path from A to F whose interior nodes avoid $\{C, D\}$. One such path is enough to say “no.”

Problem 5: Tree CPDs and context-specific independence

Reading a tree CPD. Start at the root and follow the branch that matches the observed value at each internal node until you reach a leaf. Variables that are not on that path are *irrelevant* in that context, even if you were given their values.

Worked example tree. Leaves are $(P(e^0), P(e^1))$.



- $P(e^1 \mid x^1, y^1, w^0)$: the path is $X \rightarrow x^1 \rightarrow Y \rightarrow y^1 \rightarrow W \rightarrow w^0$, so the answer is 0.9.
- $P(e^0 \mid x^0, y^1, w^1)$: the path stops right away at x^0 , so the answer is 0.4. The values of Y and W are ignored.

Context-specific independence (Lec 4, slide 16). $E \perp_c V \mid c$ means: once the context c is fixed, $P(E \mid \text{other parents}, V, c)$ is the same for every value of V , *for every setting of the remaining parents*. To check this:

1. Mark the branch for the context value in every place it appears in the tree. If the context variable appears in several places, mark all of them.
2. Look at every path that is consistent with the context. Does V appear as a split on any of them?
3. If V splits on even one consistent path, and its children give different leaves, the answer is **no**.

In the example:

- $E \perp_c Y \mid x^0$? **Yes.** Under x^0 there is only a leaf, so Y never appears.

- $E \perp_c W \mid y^0$? **Yes.** The paths consistent with y^0 are $x^0 \rightarrow \text{leaf}$ and $x^1 \rightarrow y^0 \rightarrow \text{leaf}$. W appears on neither of them.
- $E \perp_c Y \mid w^0$? **No.** W sits *below* Y . Fixing w^0 does not stop Y from mattering on the x^1 branch, where y^0 gives (0.7, 0.3) and y^1, w^0 gives (0.1, 0.9).

Tip: fixing a variable that sits *deep* in the tree, or on only one branch, usually does not remove the influence of variables that split on *other* branches. Keep this in mind for (e).